Edge-Object Co-Driven Learning for Remote Sensing Change Detection

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***Abstract*— Remote sensing change detection (CD) aims to accurately reveal surface changes by comparing two temporally separated images of the same area. However, in complex environments, insufficient edge detail recognition and limited feature extraction often affect the accuracy of CD. For this purpose, we propose a novel method named the edge-object co-driven learning network (EOCLNet), which employs a combination of the pyramid vision transformer (PVT) and the fast segment anything model (FastSAM) as parallel feature extractors to capture rich multilevel features. Specifically, it includes three key components, which are the edge extraction module (EEM), the object revelation module (ORM), and the edge-object learning (EOL). EEM explicitly captures edge details by combining low-level spatial features with high-level semantic features, providing essential edge knowledge. ORM reveals changed objects by aggregating the highest two levels of semantic features, providing initial change guidance. EOL is designed to implicitly mine edge clues by establishing relationships between edges and changed objects across multiple levels, receiving outputs from both EEM and ORM. Furthermore, during the training process, the uncertainty from the previous level's change map is utilized to guide the learning at the next level, thereby achieving a transition from uncertainty to certainty. The effectiveness of EOCLNet is validated on three public datasets, where it outperforms several state-of-the-art CD methods.**

***Index Terms*— Change detection (CD), edge extraction, edge-object learning (EOL), object revelation, remote sensing.**

I. Introduction

REMOTE sensing change detection (CD) involves the comparison of two temporally separated images of the same area to accurately identify surface changes. These changes typically include those caused by natural disasters (such as earthquakes, floods, typhoons), environmental shifts (such as deforestation, wetland degradation), or human activities (such as urban expansion, agricultural development). As one of the major technologies in the field of Earth observation, it plays a significant role in a variety of practical applications, including but not limited to land use planning [1], [2], environmental monitoring [3], [4], urban management [5], [6], and disaster assessment [7], [8]. These applications demonstrate the extensive applicability and significant value of CD technology in the real world.

At the outset of CD research, many methods emerged that primarily rely on manually extracted features to identify changes. These methods are typically divided into two types: pixel-based and object-based, together constituting traditional CD techniques [9]. Pixel-based CD methods identify changes by comparing pixel values from two distinct time points, and they fall into three main categories: algebra-based (such as image quantification [10], image difference [11]), transformation-based (such as independent component analysis [12], change vector analysis [13]), and classification-based (such as random forest [14], decision tree [15]). Although these methods are easy to implement and low in computational cost, their detection accuracy is often compromised by their high sensitivity to noise and a lack of consideration for spatial contextual information. On the other hand, object-based methods overcome the limitations of pixel-based methods by analyzing groups of pixels (termed as objects), rather than individual pixels. These methods improve detection accuracy by utilizing additional information such as shape, texture, and spatial relationships to better leverage spatial context [16]. However, object-based methods have significant flexibility in the choice of segmentation algorithms [17], [18] and parameter settings, which can lead to considerable variability in their performance across different application contexts. Overall, as remote sensing technology advances and image resolution increases, the limitations of traditional methods become increasingly apparent, urgently requiring more automated and efficient technologies to address the challenges of modern CD.

The advent of deep learning (DL) technology [19] has revolutionized CD by enabling automatic feature extraction from data, significantly enhancing automation and accuracy without reliance on expert knowledge. This technology is particularly effective for analyzing high-resolution and dynamically complex remote sensing data. Convolutional neural networks (CNNs) [20], as one of the most popular DL architectures, have well-demonstrated their powerful feature

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extraction capabilities in visual tasks [21], [22]. Fully convolutional networks (FCNs) [23] further optimize traditional CNNs by enabling end-to-end learning, becoming a fundamental basis for subsequent CD research. As a derivative of FCNs, UNet [24] utilizes a symmetric encoder–decoder structure with skip connections to enhance feature transmission, making it highly effective in CD tasks. Daudt et al. [25] developed three UNet-based network models. FC-EF concatenates image pairs to process them as a single input, while FC-Siam-Conc and FC-Siam-Diff utilize a Siamese architecture to parallel process image pairs for effective comparison of features across different time states. Due to the Siamese architecture's significant advantages in recognizing temporal changes and analyzing image similarities, it has become a popular method in current CD research. To improve the interpretability and efficiency of models, attention mechanisms have been introduced into CD networks, including spatial [26], channel [27], and self-attention [28], [29]. These mechanisms strengthen the capture of key features and effectively suppress irrelevant information. Additionally, due to the limitations of CNNs in capturing global contextual information, the proposal of Transformers [30], [31] has addressed this shortfall. Its internal self-attention mechanism enables effective handling of long-range dependencies in images. Therefore, the combination of Transformers and CNNs has also garnered widespread attention in the field of CD.

Existing CD research primarily focuses on obtaining richer image features and conducting feature fusion. This includes bitemporal feature fusion, such as our previous work, FADGNet [32], which utilizes difference information to guide the learning of fused features, and multilevel feature fusion, like AFCF3DNet [33], which designs a module (called adjacent-level feature cross-fusion) to aggregate features from adjacent levels. However, these methods often overlook the importance of edge information, which usually leads to deficiencies in edge detail recognition. Although some studies have attempted to improve this by introducing edges as an auxiliary task, such as EGDENet [34] and EGRCNN [35], these methods have not effectively established relationships between edges and changed objects, thereby limiting their effectiveness in complex scenarios. On the other hand, although DL methods like BIT [36] and USSFCNet [37] have achieved success in practical applications, their potential for further improving performance is still constrained by the insufficient features extracted from limited data.

To address the aforementioned issues, we propose a novel edge-object co-driven learning network (EOCLNet) that utilizes a combination of the pyramid vision transformer (PVT) [38] and the fast segment anything model (FastSAM) [39] as parallel feature extractors to capture rich multilevel features. Specifically, it includes three key components. The edge extraction module (EEM) is used to explicitly capture edge information, and the object revelation module (ORM) is used to reveal changed objects. Together, they provide edge knowledge and initial change guidance. The edge-object learning (EOL) process aims to implicitly mine edge clues across multiple levels of features. Furthermore, uncertainty from the change map is introduced to guide the training process, aiming to achieve a transition from uncertainty to certainty. Overall, this article primarily contributes to the following five aspects.

1) Given that FastSAM encompasses a broad range of general features, we design a dual-branch parallel feature extractor (DPFE) combining PVT and FastSAM to capture rich multilevel features.
2) EEM is constructed for explicitly acquiring edge knowledge. It first utilizes low-level spatial features to capture edge information and subsequently filters background noise by incorporating high-level semantic features.
3) Similarly, ORM is used to reveal changed objects. It aggregates the highest two levels of semantic features to capture contextual information, thereby generating initial change guidance.
4) EOL receives outputs from EEM and ORM, utilizing the acquired edge knowledge to dynamically interact with multilevel features, thereby establishing relationships between edges and changed objects to implicitly mine edge clues. Simultaneously, edge alignment loss is introduced to align edge features across different levels.
5) Furthermore, the uncertainty from the previous level's change map is utilized to construct training weights, guiding the learning at the next level to achieve a transition from uncertainty to certainty.

The remaining structure of the article is as follows. Section II reviews relevant research on CD. Section III provides a detailed introduction to the proposed EOCLNet, including the framework, key components, and loss functions. Section IV presents the experimental results and analysis. Section V discusses the proposed EOCLNet from five aspects. Finally, Section VI provides a conclusion to the article.

II. Related Works

The development of DL technology has provided new research directions for CD, prompting the proposal of many DL-based methods to address various issues. Below, we will discuss DL-based research from three aspects: DL architectures, feature fusion, and auxiliary tasks.

A. DL Architectures

CNNs, renowned for their superior feature extraction capabilities, are highly favored in the CD field, prompting the development of numerous CNN-based methods. Liu et al. [27] proposed a super-resolution CD network to address the issue of resolution inconsistency between bitemporal images. Specifically, it consists of two parts: a super-resolution module and a CD module. The former is used to unify the resolution size, while the latter outputs the CD results. Due to the fixed receptive field limitations of conventional CNNs, dilated convolutions have been introduced to expand these fields without adding parameters, enhancing feature extraction efficiency. Guo et al. [40] designed a parallel convolution structure accompanied by various dilation rates to enhance feature representation. Additionally, asymmetric convolution has been introduced to reduce parameters while preserving processing capability. Lei et al. [41] replaced standard convolutions with asymmetric convolution (a parallel combination of horizontal

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and vertical convolutions) to capture multilevel features and reduce computational cost. These CNN-based methods have made significant progress, yet they face limitations in capturing global contextual information, which hinders further improvements in performance.

The introduction of Transformers overcomes the limitations of CNNs, offering superior contextual modeling capabilities that significantly enhance the understanding of global context. In [42], a transformer-based Siamese network is proposed, which utilizes a feature extractor constructed with Transformers to capture multilevel features enriched with extensive contextual information. Considering the complementary features of Transformers and CNNs, many studies have combined them to capitalize on their synergistic benefits, typically employing either serial or parallel configurations. Chen et al. [36] employed a serial approach to combine CNNs and Transformers, with the former for feature extraction and the latter for contextual modeling of bitemporal features. In [43], a parallel feature extractor constructed with CNNs and Transformers is proposed to capture multilevel local and global features. Although the combination of CNNs and Transformers has achieved practical success, they are still constrained by the insufficiency of features extracted from limited data. Additionally, we note that visual foundation models trained on large-scale datasets, such as FastSAM [44], can provide a broader range of general features, reducing the dependency on task-specific labeled data. Meanwhile, compared to SAM [45], it is built on CNNs, which reduces computational demands. Therefore, we propose a DPFE constructed with FastSAM and PVT to mitigate the insufficiency of extracted features.

B. Feature Fusion

Feature fusion generally involves bitemporal feature fusion and multilevel feature fusion. The former is used to acquire bitemporal information, while the latter combines multilevel features to reveal CD results. Conventional methods for acquiring bitemporal information generally include feature concatenation and point-wise addition or subtraction of features. However, these methods often affect the final detection performance due to information redundancy or loss. To address this, researchers have sought more effective solutions, which can be broadly divided into two directions: designing structures that incorporate multiple bitemporal feature fusion methods and introducing attention mechanisms. Feng et al. [46] combined feature concatenation and point-wise subtraction to balance information redundancy and information loss. Lei et al. [41] constructed a difference enhancement module that utilizes point-wise subtraction to obtain difference information and enhances it through channel attention. Similarly, Zhao et al. [47] constructed channel and spatial branches to obtain more effective information before fusing bitemporal features. The above methods have achieved certain results. In this article, we combine cosine similarity with attention mechanisms to acquire bitemporal information, with detailed information provided in Section III-A.

Multilevel feature fusion aims to provide rich feature representations for revealing the final CD results. Peng et al. [48] designed a UNet++ network with four outputs, aimed at integrating change information from different levels. Due to spatial and semantic differences between features across different levels, directly combining them may obscure crucial change details, affecting the accuracy of detection. In response to the aforementioned issue, Zhang et al. [49] proposed a module (called multiscale spatial feature attention) that automatically assigns weight maps to features across different levels to facilitate their fusion. Li et al. [50] constructed a supervised attention module to achieve progressive feature fusion. Specifically, it facilitates the aggregation of features from high to low levels by reweighting features and adds supervision to alleviate the learning difficulty of features across different levels. However, most existing studies have overlooked the preservation of edge details in the process of multilevel feature fusion. To address this, we propose a feature fusion process co-driven by edges and changed objects to facilitate the learning of edge features.

C. Auxiliary Tasks

An auxiliary task refers to providing additional support for the main task of CD, typically including processes such as semantic segmentation and edge detection. Semantic segmentation generates more discriminative feature representations for CD tasks by providing additional semantic constraints. In [51], a dual-task constrained network is proposed, consisting of a CD branch and two semantic segmentation branches, which utilize the same encoder to capture multilevel features. Shen et al. [26] proposed a semantic feature-constrained CD network, which has a structure similar to the aforementioned method, utilizing semantic information to guide the generation of CD results. The semantic segmentation branches in the aforementioned methods rely on labeled data to provide semantic constraints. Chen et al. [52] adopted a self-supervised learning strategy to provide semantic constraints for CD, reducing the dependence on labeled data.

Edge detection generates more precise boundary delineations for CD tasks by providing additional edge constraints. The EGDENet [34] and EGRCNN [35] mentioned in Section I both employ edge detection as an auxiliary task to provide edge knowledge for CD. Additionally, Ma et al. [53] constructed an edge-aware decoder aimed at utilizing auxiliary edge knowledge to enhance CD results. Although the above three edge-assisted methods have achieved certain effects, they have not established a direct relationship between edges and changed objects, which limits the improvements they bring. Therefore, in this article, we propose an edge-object co-driven learning method aimed at establishing relationships between edges and changed objects across multiple levels, thereby fully mining edge clues.

III. Methodology

This section provides a detailed introduction to the proposed EOCLNet, including its framework, key components, and the training process.

A. Framework

Fig. 1 illustrates the framework of EOCLNet, which employs a DPFE to acquire rich bitemporal features.

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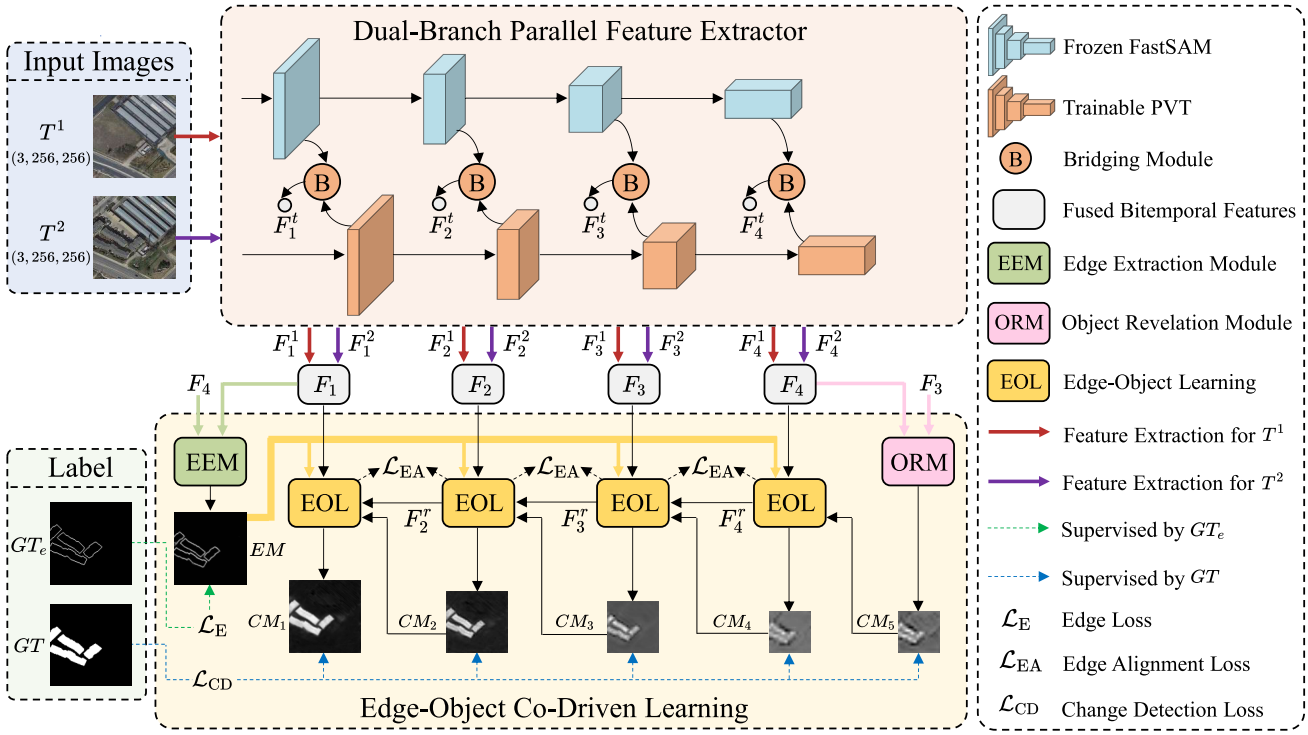


Fig. 1. Framework of the proposed EOCLNet. The DPFE captures multilevel features of bitemporal images. It includes a frozen FastSAM and a trainable PVT, with four bridging modules that combine them. Then, the bitemporal features extracted by DPFE are fused, resulting in four-level fused features. Edge knowledge is obtained by EEM (green block), which receives F_1 and F_4 as inputs. Concurrently, ORM (pink block) aggregates the highest two levels of semantic features, F_3 and F_4, to provide initial change guidance. EOL (yellow block) utilizes this information to further mine edge clues across multiple levels. Finally, the output CM_1 from the first-level EOL is used as the final predicted change map.

Specifically, the frozen FastSAM is used to capture a broad range of general features, while the trainable PVT extracts specific features for CD tasks. The bridging module is used to combine these two types of features, including a receptive field block (RFB) [54] and two learnable parameters, s_m^t and p_m^t, with the constraint that $s_m^t + p_m^t = 1$. RFB is composed of convolutions with varying receptive fields. It serves two primary functions: first, selecting general features, and second, generating features with the same number of channels. The following equation demonstrates the process of the bridging module:

$$\boldsymbol{F}_m^t = s_m^t * \mathrm{RFB}\big(\mathrm{SAM}\big(\boldsymbol{T}^t\big)\big) \oplus p_m^t * \mathrm{RFB}\big(\mathrm{PVT}\big(\boldsymbol{T}^t\big)\big) \quad (1)$$

where m represents the level index with values $m = 1, 2, 3, 4$, t represents the time dimension with values $t = 1, 2$, SAM$(\cdot)$ represents features extracted by FastSAM, PVT$(\cdot)$ represents features extracted by PVT, $\boldsymbol{T}^t$ represents bitemporal images, and $\oplus$ represents element-wise summation.

For the fusion of bitemporal features $\boldsymbol{F}_m^t$, we employ a combination of cosine similarity and attention mechanisms, with the process illustrated in Fig. 2. In detail, similarity information is obtained by calculating the cosine similarity between $\boldsymbol{F}_m^1$ and $\boldsymbol{F}_m^2$, and then the difference information $\boldsymbol{W}^r$ is derived through a reverse operation and a sigmoid function. Simultaneously, element-wise summation and subtraction are applied to $\boldsymbol{F}_m^1$ and $\boldsymbol{F}_m^2$ to derive composite and difference features, respectively. These features are then concatenated along the channel dimension, followed by processing through

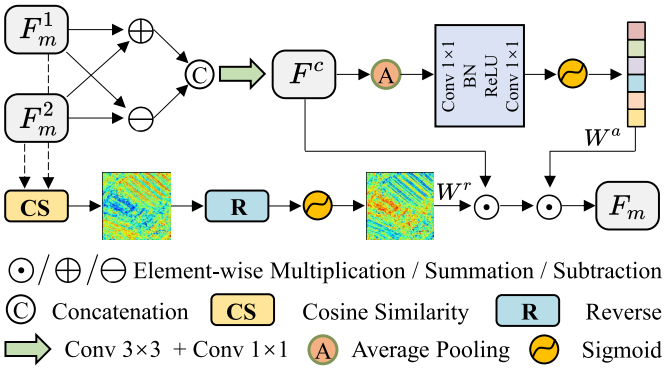


Fig. 2. Fused bitemporal features. $\boldsymbol{W}^a$ represents the channel attention weights, and $\boldsymbol{W}^r$ is derived as the difference information through cosine similarity. Together, they enhance the initially fused features $\boldsymbol{F}^c$, resulting in $\boldsymbol{F}_m$. $\boldsymbol{F}^c$ is obtained by concatenating the composite and difference features between $\boldsymbol{F}_m^1$ and $\boldsymbol{F}_m^2$.

two convolutional layers to generate the initial fused features $\boldsymbol{F}^c$. Next, the attention weights $\boldsymbol{W}^a$ are obtained by performing average pooling on $\boldsymbol{F}^c$, followed sequentially by two convolutional layers and a sigmoid function. Finally, $\boldsymbol{W}^r$ and $\boldsymbol{W}^a$ are jointly applied to $\boldsymbol{F}^c$ to obtain the enhanced fused features $\boldsymbol{F}_m$. Subsequent processes are all based on $\boldsymbol{F}_m$.

The three key components, EEM, ORM, and EOL, correspond to the green, pink, and yellow blocks in Fig. 1, respectively. EEM combines the lowest and highest level features to acquire edge knowledge, as indicated by the green arrows in Fig. 1. ORM aggregates the highest two levels of

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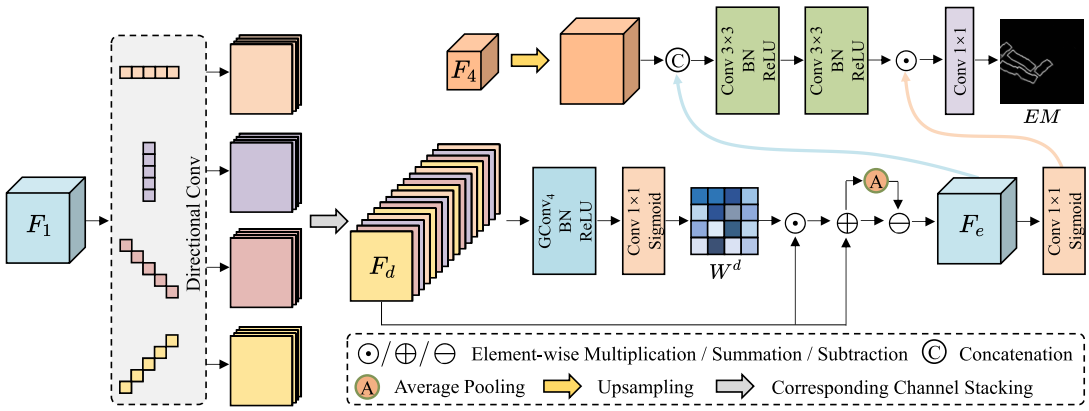


Fig. 3. Structure of EEM. Convolutions in four directions (horizontal, vertical, right diagonal, and left diagonal) are applied in parallel to $\boldsymbol{F}_1$ to construct directional weights W^d, adapting to varied directional information in bitemporal images, corresponding to (2). Then, average pooling is applied to extract edge features, corresponding to (3). Finally, $\boldsymbol{F}_4$ is combined to filter out noise interference, resulting in EM, corresponding to (4).

features to capture contextual information and provide initial change guidance, as indicated by the pink arrows in Fig. 1. EOL establishes relationships between edges and changed objects across multiple levels to implicitly mine edge clues. Additionally, it introduces edge alignment loss to align edge features across different levels. During the training phase, three types of losses are included: edge loss, edge alignment loss, and CD loss, which correspond to $\mathcal{L}_E$, $\mathcal{L}_{\text{EA}}$, and $\mathcal{L}_{\text{CD}}$ in Fig. 1, respectively. After training is completed, the output $\boldsymbol{CM}_1$ from the first-level EOL is used as the final predicted change map.

B. Edge Extraction Module

Edge information can provide effective constraints in the process of capturing changed targets, which has been confirmed in many studies [35], [53]. Considering that low-level features (such as $\boldsymbol{F}_1$ in Fig. 1) encompass texture and edge information, it is also noted that they exhibit lower robustness and are susceptible to noise interference. On the other hand, high-level features (such as $\boldsymbol{F}_4$ in Fig. 1) contain rich semantic information and possess greater robustness, but they may not be as sensitive in capturing fine textures and edge details as low-level features. Therefore, we combine the lowest and highest level features (specifically, $m = 1$ and 4) to extract edge information from $\boldsymbol{F}_1$ and employ $\boldsymbol{F}_4$ to filter out noise. This is achieved through EEM, corresponding to the green block in Fig. 1. The process of EEM is illustrated in Fig. 3.

As shown in Fig. 3, EEM can be divided into two stages: extracting edge information from $\boldsymbol{F}_1$ and combining $\boldsymbol{F}_4$ to filter out noise interference. Considering that changed objects in bitemporal images usually contain various directional information, traditional convolution may not effectively capture them. Therefore, we employ directional convolution [55] to adapt to these variations, which includes four directions: horizontal, vertical, as well as right and left diagonals. Specifically, convolutions in four directions are applied to $\boldsymbol{F}_1$ in parallel, each with a kernel size of 5 and output channels equal to one quarter of $\boldsymbol{F}_1$, as indicated by the gray dashed box in Fig. 3. Next, the features obtained from the convolutions in four directions are concatenated using corresponding channel stacking to generate $\boldsymbol{F}_d$, which has the same number of channels as $\boldsymbol{F}_1$, as indicated by the gray arrows in Fig. 3.

To facilitate interactive learning across different directions, a group convolution with a depth of 4 is performed on $\boldsymbol{F}_d$, accompanied by batch normalization and the ReLU function. Then, directional weights $\boldsymbol{W}^d$ are obtained through a 1×1 convolution layer followed by a sigmoid function.

$$\boldsymbol{W}^d = \text{sig}(\text{conv}_{1\times1}(\text{relu}(\text{bn}(\text{Gconv}_4(\boldsymbol{F}_d))))) \tag{2}$$

where $\text{sig}(\cdot)$ represents the sigmoid function, $\text{relu}(\cdot)$ represents the ReLU function, $\text{bn}(\cdot)$ represents batch normalization, $\text{conv}_{1\times1}(\cdot)$ represents a convolution with a 1×1 kernel, and $\text{Gconv}_4(\cdot)$ represents a group convolution with a 3×3 kernel and a depth of 4. $\boldsymbol{W}^d$ and $\boldsymbol{F}_d$ are multiplied element-wise, followed by summation with the original $\boldsymbol{F}_d$. Finally, average pooling is applied, and the result is subtracted from itself to extract the edge features $\boldsymbol{F}_e$

$$\boldsymbol{F}_e = \left(\boldsymbol{W}^d \odot \boldsymbol{F}_d \oplus \boldsymbol{F}_d\right) \ominus \text{avg}\left(\boldsymbol{W}^d \odot \boldsymbol{F}_d \oplus \boldsymbol{F}_d\right) \tag{3}$$

where $\text{avg}(\cdot)$ represents average pooling, and $\odot$ and $\ominus$, respectively, represent element-wise multiplication and subtraction. $\boldsymbol{F}_e$ contains crucial boundary information but is also susceptible to background noise interference. Therefore, we combine the more robust $\boldsymbol{F}_4$ to filter out these disturbances.

$\boldsymbol{F}_4$ is concatenated with $\boldsymbol{F}_e$, and subsequently passed through two convolution layers to obtain a composite feature with the same number of channels as $\boldsymbol{F}_4$ and $\boldsymbol{F}_e$. Upsampling of $\boldsymbol{F}_4$ is required before this step. Then, $\boldsymbol{F}_e$ passes through a 1×1 convolution layer followed by a sigmoid function. The output is applied to the composite feature to enhance edge information. Finally, a one-channel edge map $\boldsymbol{EM}$ is generated via another 1×1 convolution layer

$$\begin{aligned}\boldsymbol{EM} = \text{conv}_{1\times1}\big(\text{conv}^{b+r}_{3\times3}\big(\text{conv}^{b+r}_{3\times3}(\text{cat} \\ \left(\delta_{\uparrow}(\boldsymbol{F}_4), \boldsymbol{F}_e\right)\big)\big) \odot \text{sig}(\text{conv}_{1\times1}(\boldsymbol{F}_e))\big)\end{aligned} \tag{4}$$

where $\text{conv}^{b+r}_{3\times3}$ represents a convolution with a 3×3 kernel accompanied by batch normalization and the ReLU function, $\text{cat}(\cdot,\cdot)$ represents channel concatenation, and $\delta_{\uparrow}$ represents upsampling. $\boldsymbol{EM}$, as the final output of EEM, utilizes edge ground truth ($\boldsymbol{GT}_e$) for supervised learning. Additionally,

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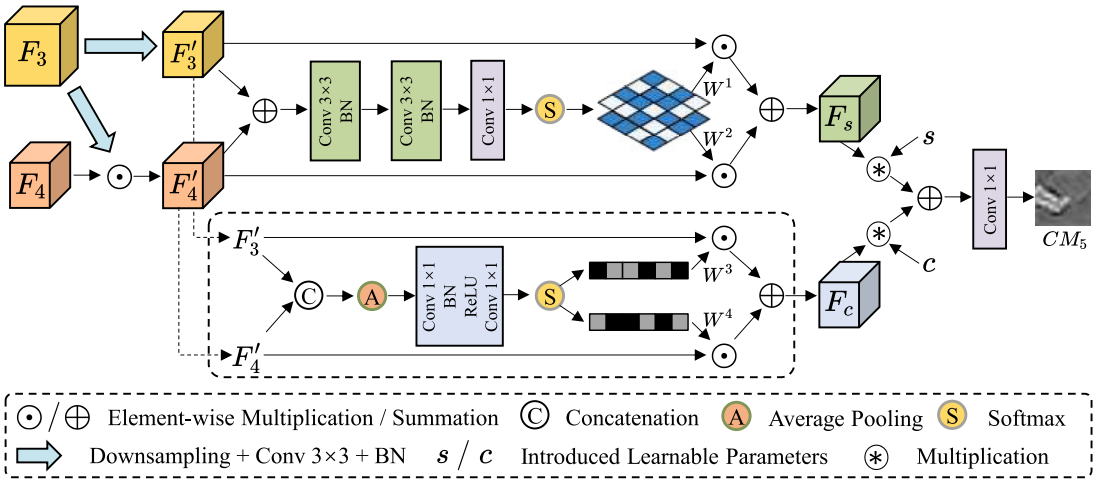


Fig. 4. Structure of ORM. Weights W^1 and W^2 are used for spatial alignment, corresponding to (5), and weights W^3 and W^4 are used for channel alignment, corresponding to (6). Prior to this, $\boldsymbol{F}_3$ and $\boldsymbol{F}_4$ are multiplied for initial information interaction. Finally, two learnable parameters, s and c, are introduced to further integrate the aligned features, resulting in $\boldsymbol{CM}_5$, which contains initial information about changed objects.

it serves as a basis for further mining edge clues across multiple levels, providing crucial edge knowledge, as indicated by the yellow arrows in Fig. 1.

C. Object Revelation Module

High-level features are rich in semantic information and robust but may lose edge details. Our expectation is to leverage the high robustness of these features to provide more distinctive information, while edge details are further mined in subsequent processes. Therefore, we aggregate the highest two levels of features (specifically, $m = 3$ and 4) to initially reveal changed objects at a lower computational cost, achieved through ORM (pink block in Fig. 1). The ORM process is illustrated in Fig. 4.

Considering the semantic differences between the highest two levels of features, we suppress these differences and align feature representations across spatial and channel dimensions. Prior to this, $\boldsymbol{F}_3$ is downsampled to match the size of $\boldsymbol{F}_4$, resulting in $\boldsymbol{F}_3'$. Additionally, the downsampled $\boldsymbol{F}_3$ is applied to $\boldsymbol{F}_4$ using element-wise multiplication to facilitate initial information interaction, yielding $\boldsymbol{F}_4'$. For spatial alignment, $\boldsymbol{F}_3'$ and $\boldsymbol{F}_4'$ are fused through element-wise summation to integrate feature information at corresponding positions. Then, the fused features sequentially pass through several convolution layers. Next, weights $\boldsymbol{W}^1$ and $\boldsymbol{W}^2$ are obtained by applying the softmax function, where the corresponding elements of $\boldsymbol{W}^1$ and $\boldsymbol{W}^2$ sum to 1. Finally, $\boldsymbol{W}^1$ and $\boldsymbol{W}^2$ are multiplied with $\boldsymbol{F}_3'$ and $\boldsymbol{F}_4'$, respectively, to achieve spatial alignment, and the results are summed to obtain $\boldsymbol{F}_s$

$$\begin{cases} \boldsymbol{W}^1, \boldsymbol{W}^2 = \text{soft}\big(\text{conv}_{1\times1}\big(\text{conv}_{3\times3}^{b}\big(\text{conv}_{3\times3}^{b}\big(\boldsymbol{F}_3' \oplus \boldsymbol{F}_4'\big)\big)\big)\big) \\ \boldsymbol{F}_s = \boldsymbol{W}^1 \odot \boldsymbol{F}_3' \oplus \boldsymbol{W}^2 \odot \boldsymbol{F}_4' \end{cases} \tag{5}$$

where soft$(\cdot)$ represents the softmax function, and $\text{conv}_{3\times3}^{b}$ represents a convolution with a 3×3 kernel accompanied by batch normalization.

For channel alignment, $\boldsymbol{F}_3'$ and $\boldsymbol{F}_4'$ are concatenated to preserve their original channel information. Then, average pooling is applied, followed by two convolution layers. The softmax function generates weights $\boldsymbol{W}^3$ and $\boldsymbol{W}^4$, whose corresponding elements also sum to 1. Finally, $\boldsymbol{W}^3$ and $\boldsymbol{W}^4$ are multiplied with $\boldsymbol{F}_3'$ and $\boldsymbol{F}_4'$, respectively, to achieve channel alignment, and the results are summed to obtain $\boldsymbol{F}_c$

$$\begin{cases} \boldsymbol{W}^3, \boldsymbol{W}^4 = \text{soft}\big(\text{conv}_{1\times1}\big(\text{conv}_{1\times1}^{b+r}\big(\text{avg}\big(\text{cat}\big(\boldsymbol{F}_3', \boldsymbol{F}_4'\big)\big)\big)\big)\big) \\ \boldsymbol{F}_c = \boldsymbol{W}^3 \odot \boldsymbol{F}_3' \oplus \boldsymbol{W}^4 \odot \boldsymbol{F}_4'. \end{cases} \tag{6}$$

To further integrate the features after spatial and channel alignment, two learnable parameters s and c are introduced, with $s + c = 1$. These parameters are multiplied with $\boldsymbol{F}_s$ and $\boldsymbol{F}_c$, respectively, and the results are summed. The sum is then passed through a 1×1 convolution layer to generate a one-channel change map $\boldsymbol{CM}_5$. As the final output of ORM, $\boldsymbol{CM}_5$ uses CD ground truth ($\boldsymbol{GT}$) for supervised learning and provides essential change information for subsequent edge clue mining.

D. Edge-Object Learning

As previously mentioned, $\boldsymbol{EM}$ and $\boldsymbol{CM}_5$ derived from EEM and ORM, respectively, provide edge knowledge and initial change guidance. The proposed EOL process utilizes $\boldsymbol{EM}$ to dynamically interact with multilevel features, thereby establishing relationships between edges and changed objects. In this process, edge clues are implicitly mined, and changed objects are progressively refined, corresponding to the yellow block in Fig. 1. The process of EOL is displayed in Fig. 5.

The mth level EOL generally includes four inputs: the fused bitemporal features $\boldsymbol{F}_m$ and the edge map $\boldsymbol{EM}$ derived from Figs. 2 and 3, as well as the refined features $\boldsymbol{F}_{m+1}^r$ and the change map $\boldsymbol{CM}_{m+1}$ from the outputs of a higher-level EOL. A special note is made that the highest-level EOL (specifically, $m = 4$) includes three inputs: the previously mentioned $\boldsymbol{F}_m$

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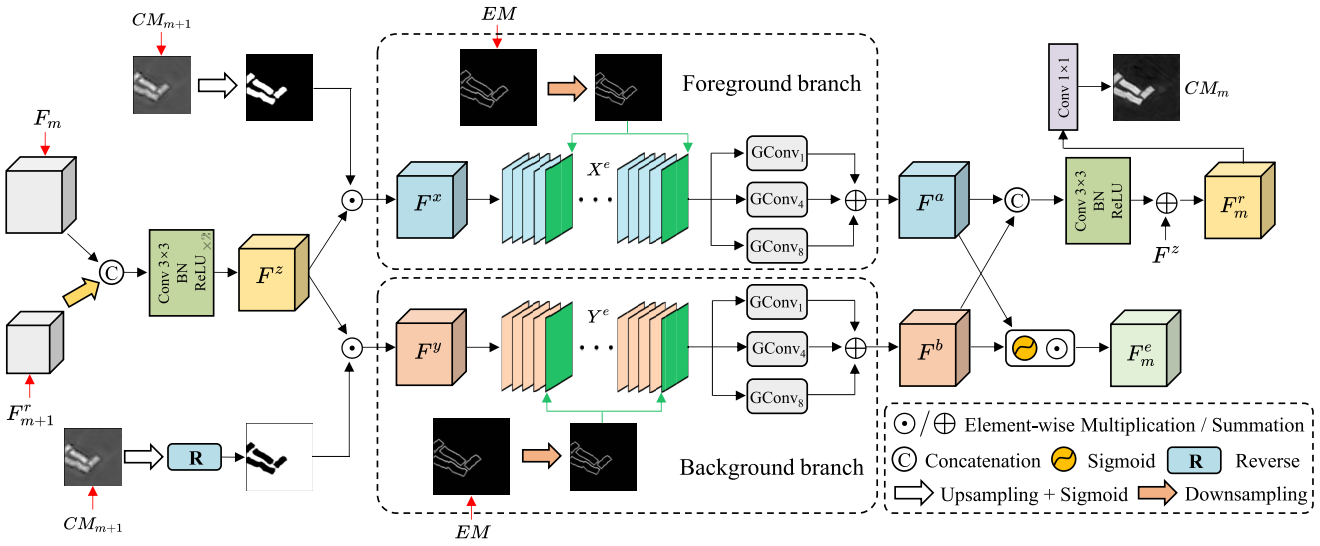


Fig. 5. Process of EOL. The inputs include $\boldsymbol{F}_m$, $\boldsymbol{EM}$, $\boldsymbol{F}^r_{m+1}$, and $\boldsymbol{CM}_{m+1}$, with the first two derived from Figs. 2 and 3, and the last two from the outputs of a higher-level EOL. $\boldsymbol{F}^r_{m+1}$ and $\boldsymbol{F}_m$ are concatenated to increase the diversity of change features. $\boldsymbol{CM}_{m+1}$ is used to segregate foreground and background features. EM is involved in the construction of both foreground and background branches, providing edge knowledge. The outputs are denoted as $\boldsymbol{F}^e_m$, $\boldsymbol{F}^r_m$, and $\boldsymbol{CM}_m$.

and $\boldsymbol{EM}$, as well as the initial change map $\boldsymbol{CM}_5$ derived from Fig. 4, which can be observed in Fig. 1. The outputs of the mth level EOL are denoted as $\boldsymbol{F}^e_m$, $\boldsymbol{F}^r_m$, and $\boldsymbol{CM}_m$. $\boldsymbol{F}^e_m$ is used for edge alignment, and $\boldsymbol{F}^r_m$ and $\boldsymbol{CM}_m$ serve as inputs for lower-level EOL.

To establish relationships between edges and changed objects, we construct a foreground branch and a background branch. As a preparatory step, the refined features $\boldsymbol{F}^r_{m+1}$ from the higher-level EOL output are upsampled and concatenated with $\boldsymbol{F}_m$ to enrich feature diversity. Subsequently, $\boldsymbol{F}^z$ is obtained after passing through multiple convolution layers. For the foreground branch, the initial step is to obtain the foreground feature $\boldsymbol{F}^x$ using $\boldsymbol{CM}_{m+1}$. Specifically, this is done by upsampling $\boldsymbol{CM}_{m+1}$ and multiplying it with $\boldsymbol{F}^z$. To mine edge clues in $\boldsymbol{F}^x$, we utilize the edge map $\boldsymbol{EM}$ derived from Fig. 3 to provide edge knowledge. Specifically, $\boldsymbol{F}^x$ is divided into k groups along the channel dimension, where $k = 8$. Simultaneously, $\boldsymbol{EM}$ is downsampled to match the size of $\boldsymbol{F}^x$ and replicated k times. Each copy of $\boldsymbol{EM}$ is then inserted into a corresponding group, and these k groups are concatenated to form $\boldsymbol{X}^e$, corresponding to the green arrows in Fig. 5. Finally, group convolutions of varying depths are applied in parallel to $\boldsymbol{X}^e$ to facilitate comprehensive learning, and their results are summed to obtain $\boldsymbol{F}^a$

$$\begin{cases} \boldsymbol{F}^x \to \{\boldsymbol{f}^x_1, \boldsymbol{f}^x_2, \ldots, \boldsymbol{f}^x_k\} \\ \boldsymbol{X}^e = \text{cat}(\boldsymbol{f}^x_1, \boldsymbol{EM}, \boldsymbol{f}^x_2, \boldsymbol{EM}, \ldots, \boldsymbol{f}^x_k, \boldsymbol{EM}) \\ \boldsymbol{F}^a = \text{Gconv}_1(\boldsymbol{X}^e) \oplus \text{Gconv}_4(\boldsymbol{X}^e) \oplus \text{Gconv}_8(\boldsymbol{X}^e) \end{cases} \tag{7}$$

where Gconv_i represents a group convolution with a 3×3 kernel and depths of $i = 1, 4, 8$. For the background branch, the background feature $\boldsymbol{F}^y$ is obtained by reversing $\boldsymbol{CM}_{m+1}$ and then multiplying it with $\boldsymbol{F}^z$. Clearly, $\boldsymbol{CM}_{m+1}$ needs to be upsampled beforehand to match the size of $\boldsymbol{F}^z$. The process of mining edge clues in $\boldsymbol{F}^y$ is similar to that of the foreground branch, ultimately leading to the output of the background branch, $\boldsymbol{F}^b$.

The outputs from the foreground and background branches, $\boldsymbol{F}^a$ and $\boldsymbol{F}^b$, are rich in edge clues. They are processed with the sigmoid function, and their results are multiplied to extract edge features, yielding $\boldsymbol{F}^e_m$. It is utilized to construct edge alignment loss, aligning edge features across different levels of EOL outputs. Additionally, $\boldsymbol{F}^a$ and $\boldsymbol{F}^b$ are concatenated, then passed through a 3×3 convolution layer and summed with the original $\boldsymbol{F}^z$ to produce the refined feature $\boldsymbol{F}^r_m$. $\boldsymbol{F}^r_m$ is further processed through a 1×1 convolution layer to produce the change map $\boldsymbol{CM}_m$. Both $\boldsymbol{F}^r_m$ and $\boldsymbol{CM}_m$ serve as inputs for lower-level EOL, and $\boldsymbol{CM}_m$ is supervised using $\boldsymbol{GT}$.

In the proposed EOCLNet, EOL is employed after each $\boldsymbol{F}_m$ to establish relationships between edges and changed objects across different levels of features, thereby implicitly mining edge clues as shown by the yellow block in Fig. 1. This differs from previous methods mentioned that use edges as an auxiliary task [34], [35], which merely construct an edge supervision branch to explicitly increase edge constraints. Our method combines the explicit learning of EEM with the implicit mining process of EOL.

E. Training Process

The proposed EOCLNet includes the main task of CD, as well as explicit learning and implicit mining processes for edges. Therefore, during the training phase, there are three types of losses: CD loss $\mathcal{L}_{\text{CD}}$, edge alignment loss $\mathcal{L}_{\text{EA}}$, and edge loss $\mathcal{L}_E$.

The $\mathcal{L}_{\text{CD}}$ corresponds to the change maps output by ORM and EOL. Considering the complexity of change scenarios, there is often ambiguity in the CD results. This ambiguity stems from the uncertainty in pixel attribute determinations. Specifically, the generated change maps are processed through a sigmoid function, ensuring pixel values range from 0 to 1,

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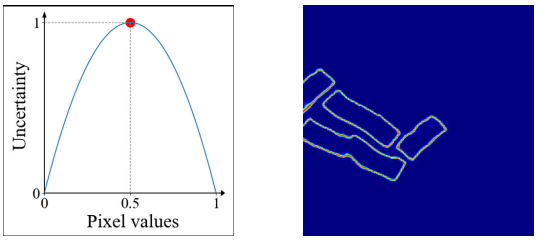


Fig. 6. Uncertainty illustration. (a) Uncertainty measure. (b) Uncertainty map $\boldsymbol{U}_m$.

where 1 indicates changed pixels and 0 indicates unchanged pixels. Clearly, the closer the pixel values are to 1 or 0, the greater the certainty (or the lesser the uncertainty) of change or no change, respectively. Conversely, values closer to 0.5 indicate greater uncertainty, with the maximum uncertainty occurring at a pixel value of 0.5, as shown in Fig. 6(a). Therefore, during training, it is necessary to focus more on areas of uncertainty (i.e., pixel values close to 0.5) to facilitate the transition from uncertainty to certainty. To achieve this, we construct a weight map $\boldsymbol{\Omega}_m$ for the binary cross-entropy (BCE) loss, aiming to impose greater penalties on pixel values close to 0.5 during training. The calculation of $\boldsymbol{\Omega}_m$ is as follows:

$$\begin{cases} \boldsymbol{U}_m = 1 - |2 * \text{sig}(\boldsymbol{CM}_m) - 1|^2 \\ \boldsymbol{\Omega}_m = 1 + \alpha * \boldsymbol{U}_m \end{cases} \tag{8}$$

where $|\cdot|$ represents absolute value, $\boldsymbol{U}_m$ represents an uncertainty map, $\boldsymbol{\Omega}_m$ represents the weight map constructed based on uncertainty, and α is the scaling factor, set to 3. Fig. 6(b) displays the uncertainty map $\boldsymbol{U}_m$, where it can be observed that the areas of uncertainty are primarily concentrated near the edges. Therefore, increasing focus on areas of uncertainty during training can further enhance the learning of edges, facilitating the generation of clearer boundaries.

Additionally, given the imbalance between changed and unchanged classes, the dice (Dice) loss is also introduced as part of the CD loss. The $\mathcal{L}_{\text{CD}}$ is summarized as follows:

$$\mathcal{L}_{\text{CD}} = \sum_{m=1}^{5} \lambda \Big(\text{BCE}\big(\delta_{\uparrow}(\text{sig}(\boldsymbol{CM}_m)), \boldsymbol{GT}, \delta_{\uparrow}(\boldsymbol{\Omega}_{m+1})\big) + \text{Dice}\big(\delta_{\uparrow}(\text{sig}(\boldsymbol{CM}_m)), \boldsymbol{GT}\big) \Big) \tag{9}$$

where λ is the adjustment factor, set to 0.2. For $\mathcal{L}_{\text{CD}}$, it is important to note that we use $\boldsymbol{\Omega}_{m+1}$, generated from the previous level $\boldsymbol{CM}_{m+1}$, as the training weight for the next level $\boldsymbol{CM}_m$, to guide subsequent training to focus on areas of uncertainty in the previous level change map. Additionally, when $m = 5$, $\boldsymbol{\Omega}_{m+1} = \boldsymbol{\Omega}_m$.

The $\mathcal{L}_{\text{EA}}$ corresponds to the edge features output by EOL. It utilizes the mean squared error (mse) loss to align the edge features from two adjacent levels of EOL outputs

$$\mathcal{L}_{\text{EA}} = \sum_{m=1}^{3} \text{mse}\big(\boldsymbol{F}^e_m, \delta_{\uparrow}(\boldsymbol{F}^e_{m+1})\big). \tag{10}$$

Edge features $\boldsymbol{F}^e_{m+1}$ from a higher-level EOL have lower resolution, so upsampling is necessary. The $\mathcal{L}_E$ corresponds to the edge map output by EEM. It utilizes the Dice loss

$$\mathcal{L}_E = \text{Dice}\left(\delta_{\uparrow}(\boldsymbol{EM}), \boldsymbol{GT}_e\right). \tag{11}$$

The total training loss $\mathcal{L}$ is the sum of the aforementioned three types of losses

$$\mathcal{L} = \mathcal{L}_{\text{CD}} + \mathcal{L}_{\text{EA}} + \mathcal{L}_E. \tag{12}$$

IV. Experiments

This section first presents three public CD datasets, ten comparison methods, and five evaluation metrics. Next, we briefly describe the experiment's implementation details. Finally, the effectiveness of EOCLNet is verified through comparison and ablation experiments.

A. Datasets

In the experiment, we utilize three popular datasets for remote sensing CD tasks. These include two datasets focusing on changes in the building category, namely LEVIR-CD [56] and WHU-CD [6], and one dataset covering multiple categories of changes, namely SYSU-CD [57]. Below, we provide a brief introduction to these three datasets.

1) LEVIR-CD Dataset [56]: This dataset focuses on building changes, including building growth and decline, covering a variety of building types such as tall apartments and large warehouses. It contains 637 pairs of high-resolution images, each with a spatial resolution of 0.5 m and a size of 1024 × 1024 pixels. These images document the evolution of land use in 20 different regions, including Austin and Manor in Texas, USA, spanning from 2002 to 2018. The LEVIR-CD dataset provides official division criteria, but considering GPU memory limitations, we crop each 1024 × 1024 pixel image. Specifically, it is segmented into 16 nonoverlapping patches, each patch being 256 × 256 pixels. This increases the number of training samples to 7120 pairs, while the validation and test samples contain 1024 and 2048 pairs, respectively.

2) WHU-CD Dataset [6]: This dataset documents the building changes in Christchurch, New Zealand, following the earthquake, in the years 2012 and 2016. It contains a pair of high-resolution aerial images, each with a spatial resolution of 0.2 m and a size of 15354 × 32507 pixels. Since the WHU-CD dataset does not provide official criteria for data division, we adopt the division method set by BIT [36] in our experiments. Specifically, each 15354 × 32507 pixel image is segmented into nonoverlapping patches, resulting in 7620 pairs of 256 × 256 pixel images. These are then randomly divided, with 6095 pairs designated for training, 762 for validation, and 763 for testing.

3) SYSU-CD Dataset [57]: Unlike datasets primarily focused on buildings, the SYSU-CD dataset encompasses a wider range of change types, including building transformations, road expansions, sea construction, and vegetation changes, which pose greater challenges for CD algorithms. It contains 20 000 pairs of aerial images, each with a resolution of 0.5 m and a size of 256 × 256 pixels. These images were captured in Hong Kong, spanning from 2007 to 2014. We adhere to the official division criteria set by the SYSU-CD dataset, with 12 000 pairs used for training, 4000 for validation, and 4000 for testing.

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B. Comparative Methods

To assess the effectiveness of EOCLNet, we compare it against ten representative CD methods. These methods include some classic and recent studies, namely FC-Siam-Conc [25], FC-Siam-Diff [25], DSAMNet [57], MSCANet [58], BIT [36], DDCNN [59], DMINet [46], AFCF3DNet [33], EATDer [53], and ScratchFormer [60]. A brief overview of each method is provided below.

1) FC-Siam-Conc [25]: A fully convolutional Siamese network that utilizes a shared-weight structure to capture bitemporal information and fuses them through feature concatenation.

2) FC-Siam-Diff [25]: Similar to FC-Siam-Conc, this network is also a fully convolutional Siamese structure, but it differs by fusing bitemporal information through feature differences.

3) DSAMNet [57]: An attention metric network that constructs a metric module with channel and spatial attention to learn change maps and introduces deep supervision as an auxiliary to learn more useful features.

4) MSCANet [58]: A multiscale context aggregation network that utilizes Transformers to encode and aggregate contextual information, compensating for the limitations of CNNs in capturing long-range dependencies, and designs multiple prediction heads to enhance feature learning.

5) BIT [36]: A transformer-based network that employs a bitemporal image transformer to perform contextual modeling of high-level semantic features extracted by ResNet-18, compensating for the shortcomings of CNNs in capturing long-range dependencies.

6) DDCNN [59]: A UNet-based network that embeds dense attention and difference enhancement units, with the former used for aggregating features from higher to lower levels and the latter for fully excavating change information.

7) DMINet [46]: A dual-branch multilevel intertemporal network that promotes feature interaction between temporal images through joint attention mechanisms and accurately detects changes using an incremental aggregation strategy.

8) AFCF3DNet [33]: An adjacent-level feature fusion network that employs a 3-D convolution-based fusion strategy to overcome the semantic gap between features of different levels.

9) EATDer [53]: An edge-assisted adaptive transformer network that utilizes a self-adaption visual transformer to construct the encoder, and introduces edge learning as an auxiliary task to generate change maps with clear edges.

10) ScratchFormer [60]: A transformer network trained from scratch that employs a shuffled sparse-attention to replace the conventional self-attention, designed to capture the inherent features of bitemporal images.

C. Evaluation Metrics

To more intuitively compare the performance of CD algorithms, we employ five evaluation metrics in our experimental analysis, namely precision (Pre), recall (Rec), kappa coefficient (Kappa), $F1$-score ($F1$), and intersection over union (IoU). Their expressions are as follows:

$$\begin{cases} \text{Pre} = \dfrac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Rec} = \dfrac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{Kappa} = \dfrac{\text{OA}-P}{1-P} \\ F1 = \dfrac{2\ \text{Pre}\cdot\text{Rec}}{\text{Pre}+\text{Rec}} \\ \text{IoU} = \dfrac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \end{cases} \tag{13}$$

where TP and FP, respectively, represent the number of true positives and false positives, and TN and FN, respectively, represent the number of true negatives and false negatives. In the Kappa metric, OA and P refer to the overall accuracy and the probability of random agreement, respectively. The calculations for OA and P are as follows:

$$\begin{cases} \text{OA} = \dfrac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \\ P = \dfrac{(\text{TP}+\text{FP})\cdot(\text{TP}+\text{FN})+(\text{FN}+\text{TN})\cdot(\text{FP}+\text{TN})}{(\text{TP}+\text{TN}+\text{FP}+\text{FN})^2}. \end{cases} \tag{14}$$

Given that Pre and Rec focus on specific aspects of model performance, while Kappa, $F1$, and IoU provide a more comprehensive evaluation, the higher these three metrics score, the greater the model's accuracy. Therefore, in our experimental analysis, we pay more attention to the values of Kappa, $F1$, and IoU.

D. Implementation Details

The proposed EOCLNet is developed using the PyTorch framework and utilizes the Adam optimizer [61] for parameter optimization, with an initial learning rate (lr_init) set at $1e^{-4}$. A cosine annealing strategy is employed to dynamically adjust the learning rate based on training progress. Specifically, the new learning rate (lr_new) is updated at each training epoch as follows:

$$\text{lr_new} = \text{lr_min} + \frac{1}{2}(\text{lr_init} - \text{lr_min}) * \left(1 + \cos\left(\frac{\pi * \text{epoch}}{\text{epoch_total}}\right)\right) \tag{15}$$

where lr_min is the minimum learning rate at the end of training, set to $1e^{-5}$, $\cos(\cdot)$ represents the cosine function, epoch represents the current training epoch, and epoch_total represents the total number of training epochs. Additionally, various data augmentation techniques, including random cropping and rotation, are applied during model training to enhance the model's generalizability. Upon completing each training epoch, IoU is calculated on the validation set, and the parameters currently holding the highest IoU are retained. All experiments are conducted on a device equipped with an NVIDIA GeForce RTX 3090 GPU, with batch sizes set to 8 for training and 16 for validation.

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TABLE I

QUANTITATIVE RESULTS ON THREE PUBLIC DATASETS, WITH ALL RESULTS EXPRESSED AS PERCENTAGES (%). ADDITIONALLY, THE HIGHEST VALUE IS HIGHLIGHTED IN BOLD, WHILE THE SECOND HIGHEST IS UNDERLINED

Methods	LEVIR-CD					WHU-CD					SYSU-CD				
	Pre	Rec	Kappa	$F1$	IoU	Pre	Rec	Kappa	$F1$	IoU	Pre	Rec	Kappa	$F1$	IoU
FC-Siam-Conc	89.14	84.66	86.16	86.84	76.75	75.40	92.60	82.29	83.12	71.12	80.44	72.92	69.70	76.49	61.94
FC-Siam-Diff	90.05	83.16	85.77	86.47	76.16	80.76	91.16	84.97	85.65	74.89	**90.53**	58.18	64.24	70.84	54.84
DSAMNet	80.23	89.09	83.54	84.43	73.05	81.58	78.94	79.37	80.23	66.99	74.88	82.16	71.26	78.35	64.41
MSCANet	90.41	89.10	89.20	89.75	81.40	88.20	91.99	89.60	90.05	81.81	81.33	75.70	72.06	78.42	64.49
BIT	92.16	86.98	88.95	89.50	80.99	93.59	92.92	92.96	93.26	87.37	81.90	76.19	72.74	78.94	65.21
DDCNN	92.18	89.08	90.10	90.60	82.82	96.34	87.93	91.60	91.95	85.09	81.82	78.88	74.40	80.32	67.12
DMINet	92.64	89.00	90.30	90.78	83.12	87.02	89.19	87.55	88.09	78.72	84.13	79.76	76.49	81.89	69.33
AFCF3DNet	91.90	89.80	90.35	90.84	83.21	93.82	93.12	93.18	93.47	87.74	86.47	79.98	78.15	83.10	71.08
EATDer	91.78	90.51	90.67	91.14	83.73	89.77	92.02	90.47	90.88	83.28	81.61	82.77	76.64	82.19	69.76
ScratchFormer	92.25	89.98	90.63	91.10	83.66	94.76	89.91	91.94	92.28	85.66	79.52	71.55	68.22	75.32	60.41
Ours	**93.30**	**92.02**	**92.26**	**92.66**	**86.32**	**97.28**	**94.64**	**95.76**	**95.94**	**92.20**	83.79	**85.53**	**79.85**	**84.65**	**73.39**

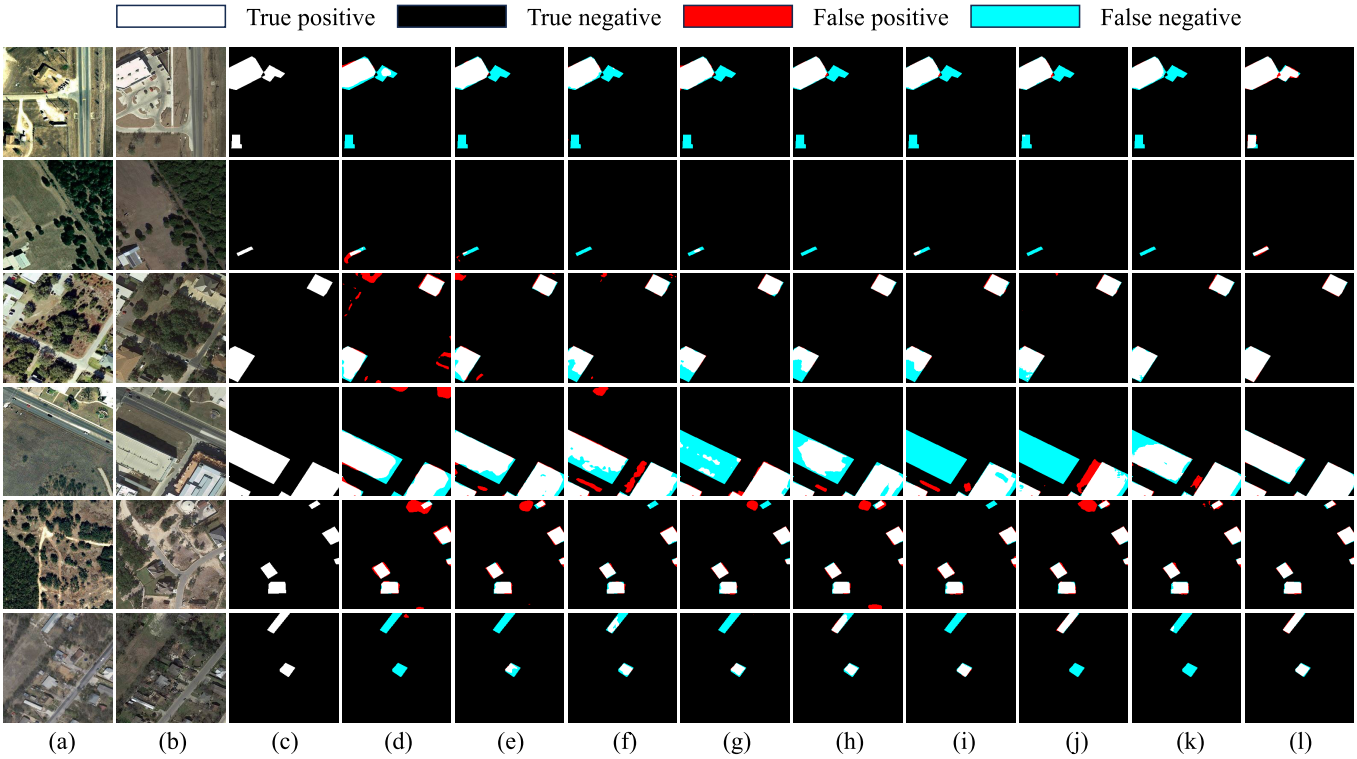


Fig. 7. Qualitative results on the LEVIR-CD dataset. (a) $\boldsymbol{T}^1$. (b) $\boldsymbol{T}^2$. (c) $\boldsymbol{GT}$. (d) DSAMNet. (e) MSCANet. (f) BIT. (g) DDCNN. (h) DMINet. (i) AFCF3DNet. (j) EATDer. (k) ScratchFormer. (l) Ours.

E. Comparative Experiments

Different CD methods are comprehensively compared from both quantitative and qualitative perspectives. Quantitative results are presented in Table I, and qualitative results can be found in Figs. 7–9.

1) Quantitative Comparison: Table I lists the quantitative results on three public datasets, clearly showing that our proposed method achieved optimal results. On the LEVIR-CD dataset, compared to EATDer, which uses edges as an auxiliary task, our method achieves higher scores in Kappa, $F1$, and IoU by 1.59%, 1.52%, and 2.59%, respectively. This validates the effectiveness of further mining edge clues by establishing relationships between edges and changed objects. ScratchFormer's performance on the LEVIR-CD dataset is second only to our method and EATDer, with its Kappa, $F1$, and IoU scores being 90.63%, 91.10%, and 83.66%, respectively. On the WHU-CD dataset, our method achieves the highest values across all five evaluation metrics, with Kappa, $F1$, and IoU scores of 95.76%, 95.94%, and 92.20%, respectively. These scores surpass the second-ranked AFCF3DNet by 2.58%, 2.47%, and 4.46%, and the third-ranked BIT by 2.80%, 2.68%, and 4.83%. Similarly, on the SYSU-CD dataset, which involves multiple types of changes, our method significantly outperforms other methods with a Kappa of 79.85%, $F1$ of 84.65%, and IoU of 73.39%. AFCF3DNet is the second highest in these three metrics. EATDer has the second highest Rec at 82.77%.

2) Qualitative Comparison: For each public dataset, six representative scenarios are selected for qualitative comparison. Additionally, to facilitate observation, different colors are used to represent TP (white), TN (black), FP (red), and FN

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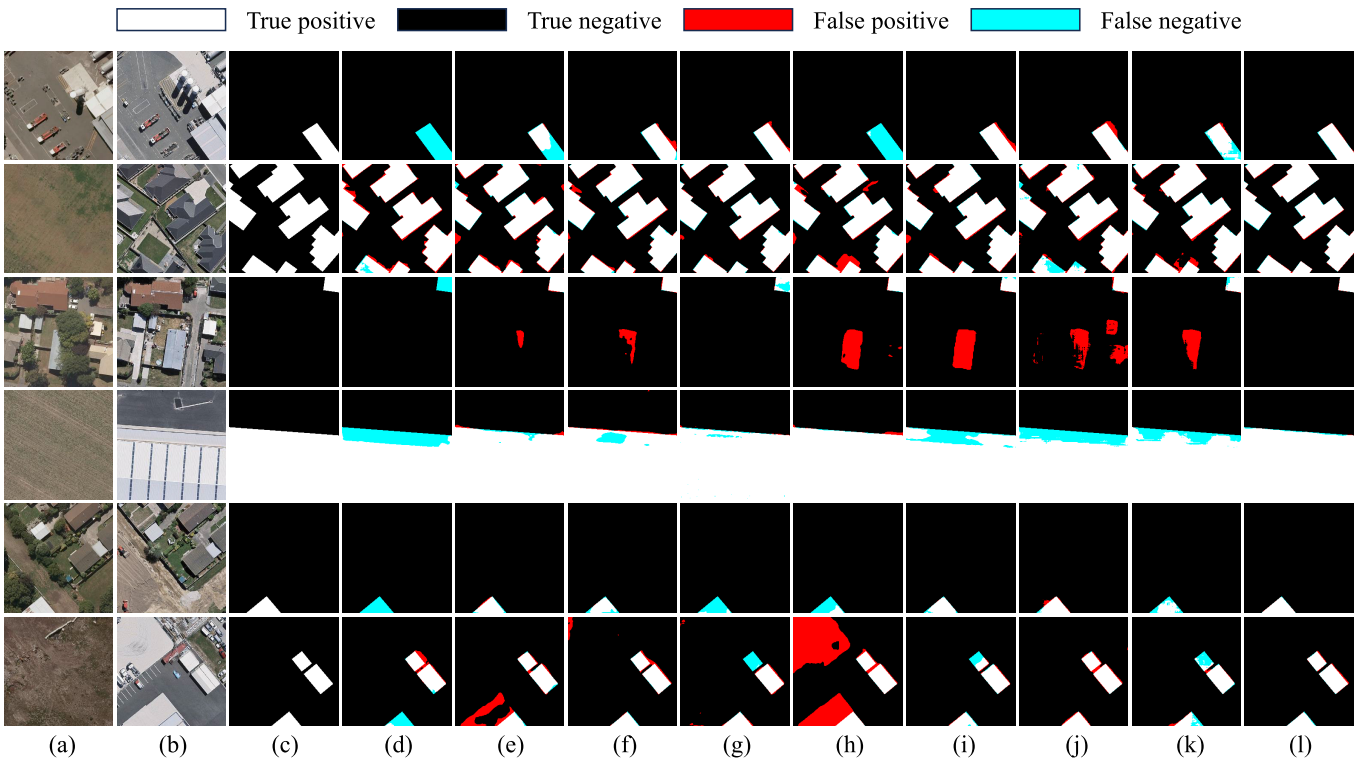


Fig. 8. Qualitative results on the WHU-CD dataset. (a) $\boldsymbol{T}^1$. (b) $\boldsymbol{T}^2$. (c) $\boldsymbol{GT}$. (d) DSAMNet. (e) MSCANet. (f) BIT. (g) DDCNN. (h) DMINet. (i) AFCF3DNet. (j) EATDer. (k) ScratchFormer. (l) Ours.

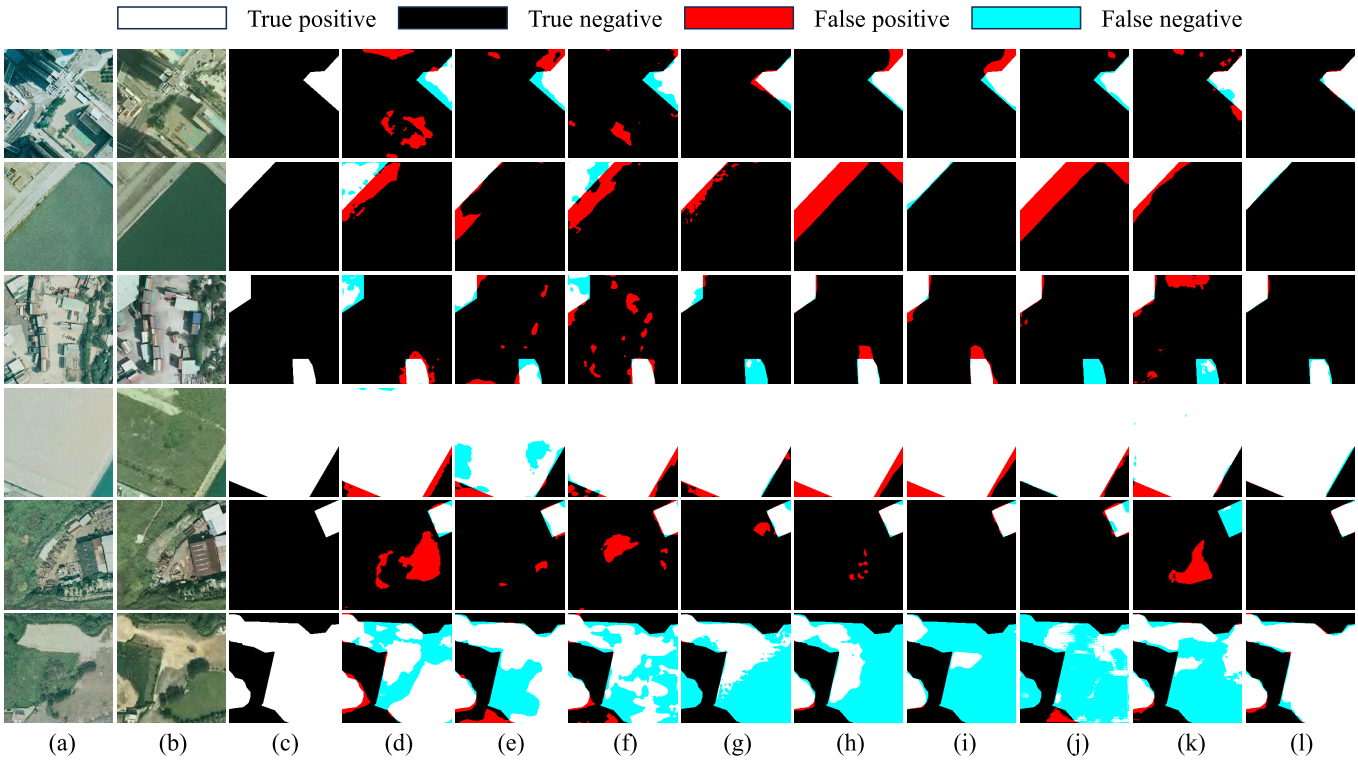


Fig. 9. Qualitative results on the SYSU-CD dataset. (a) $\boldsymbol{T}^1$. (b) $\boldsymbol{T}^2$. (c) $\boldsymbol{GT}$. (d) DSAMNet. (e) MSCANet. (f) BIT. (g) DDCNN. (h) DMINet. (i) AFCF3DNet. (j) EATDer. (k) ScratchFormer. (l) Ours.

(blue). Figs. 7–9, respectively, display the qualitative results on three public datasets. It can be observed that the proposed method achieves superior results, primarily manifested in four aspects: preservation of edge details, identification of pseudo-changes, detection of small objects, and integrity of changed objects.

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TABLE II

ABLATION STUDIES ON THREE PUBLIC DATASETS. THE UNITS FOR PARAMS ARE 10^6 (M), FLOPS ARE 10^9 (G), AND KAPPA, $F1$, AND IOU ARE EXPRESSED IN PERCENTAGES (%)

No.	Model	Efficiency		LEVIR-CD			WHU-CD			SYSU-CD		
		Params	FLOPs	Kappa	$F1$	IoU	Kappa	$F1$	IoU	Kappa	$F1$	IoU
#01	EOCLNet	27.32	56.10	92.26	92.66	86.32	95.76	95.94	92.20	79.85	84.65	73.39
(a) Dual-Branch Parallel Feature Extractor (DPFE)												
#02	w/o FastSAM	26.65	50.25	92.00	92.40	85.88	94.90	95.11	90.68	79.40	84.24	72.77
(b) Edge Extraction Module (EEM)												
#03	w/o EEM	27.26	55.06	91.91	92.32	85.73	94.64	94.87	90.24	79.18	83.90	72.27
#04	w/o $\boldsymbol{F}_4$	27.26	55.19	92.08	92.48	86.02	94.97	95.18	90.80	79.57	84.15	72.63
#05	$\{\boldsymbol{F}_1, \boldsymbol{F}_3\}$	27.32	56.10	92.17	92.57	86.16	95.60	95.79	91.92	79.74	84.36	72.95
(c) Object Revelation Modul (ORM)												
#06	w/o ORM	27.28	56.10	91.88	92.29	85.68	94.36	94.59	89.74	78.84	83.62	71.86
#07	w/o spatial alignment	27.30	56.10	92.06	92.46	85.98	95.25	95.45	91.30	79.24	83.92	72.30
#08	w/o channel alignment	27.32	56.10	92.15	92.55	86.13	95.48	95.67	91.70	79.33	84.06	72.51
#09	$\{\boldsymbol{F}_2, \boldsymbol{F}_4\}$	27.32	56.10	92.10	92.50	86.05	95.21	95.42	91.23	79.34	84.08	72.54
(d) Edge-Object Learning (EOL)												
#10	w/o EOL	26.87	53.39	91.63	92.05	85.28	94.85	95.07	90.60	78.42	83.19	71.22
#11	w/o foreground branch	27.14	55.14	92.11	92.51	86.06	95.52	95.71	91.77	79.73	84.38	72.99
#12	w/o background branch	27.14	55.14	92.06	92.46	85.97	95.31	95.50	91.39	79.69	84.27	72.82
(e) Uncertainty Weight Map ($\boldsymbol{\Omega}_m$)												
#13	w/o $\boldsymbol{\Omega}_m$	27.32	56.10	92.02	92.42	85.91	94.97	95.18	90.81	78.95	83.81	72.13

1) *Preservation of edge details:* As observed in the first row of Fig. 8, the proposed EOCLNet achieves clearer edges, while other methods, including EATDer with edge constraints, are insufficient in preserving edge details. DSAMNet and DMINet fail to detect building changes altogether. MSCANet and ScratchFormer produce incomplete change regions. Similarly, in the fourth row of Fig 9, the edge results generated by EOCLNet are the best, thanks to its further mining of edge clues across multiple levels.
2) *Identification of pseudo-changes:* In the third row of Fig. 8, pseudo-changes occur due to the obstruction of buildings by trees. Most methods, such as DMINet and AFCF3DNet, incorrectly identify buildings obscured by trees as changes, while our method does not. As observed in the fifth row of Fig. 7, many methods, such as DSAMNet and DDCNN, misclassify other types of changes as building changes. BIT and AFCF3DNet fail to detect changes in small buildings. In contrast, our method can effectively differentiate these changes.
3) *Detection of small objects:* As observed in the first row of Fig. 7, all methods can relatively completely capture newly constructed large buildings. However, for small buildings with a color similar to the background, all methods except the proposed one show poor performance. The results shown in the second row of Fig. 7 further verify that the proposed method achieves good performance in detecting changes in small objects.
4) *Integrity of changed objects:* Obtaining complete change information in complex scenarios places higher demands on CD algorithms. As shown in the sixth row of Fig. 9, only our method achieves relatively complete CD results, while other methods clearly fall short in maintaining the integrity of changed objects. This can be attributed to two main reasons. First, the constructed DPFE compensates for the insufficient features extracted from the limited dataset. Second, the introduction of the EOL process across multilevel features allows for the progressive refinement of changed objects.

F. Ablation Studies

Twelve sets of ablation experiments are conducted on three public datasets to assess the effectiveness of the key components and configurations of EOCLNet. Additionally, we provide the number of parameters (Params) and floating-point operations (FLOPs). Table II presents the results of the ablation study.

1) Effectiveness of DPFE: The constructed DPFE utilizes the general features obtained by FastSAM as a complement to the specific features. To validate its effectiveness, we use PVT only to construct the feature extractor (referred to as w/o FastSAM). The results on three public datasets are shown in #02 of Table II. On the LEVIR-CD dataset, the Kappa, $F1$, and IoU scores for w/o FastSAM are 0.26%, 0.26%, and 0.44% lower than those of EOCLNet. On the WHU-CD dataset, the decreases are 0.86%, 0.83%, and 1.52%, respectively. On the SYSU-CD dataset, the decreases are 0.45%, 0.41%, and 0.62%, respectively. This suggests that combining FastSAM and PVT as parallel feature extractors is effective.

2) Effectiveness of EEM: By combining the lowest-level and highest-level features, EEM is used to explicitly learn edge information. We replace EEM with a 1×1 convolutional layer (referred to as w/o EEM) to validate its effectiveness.

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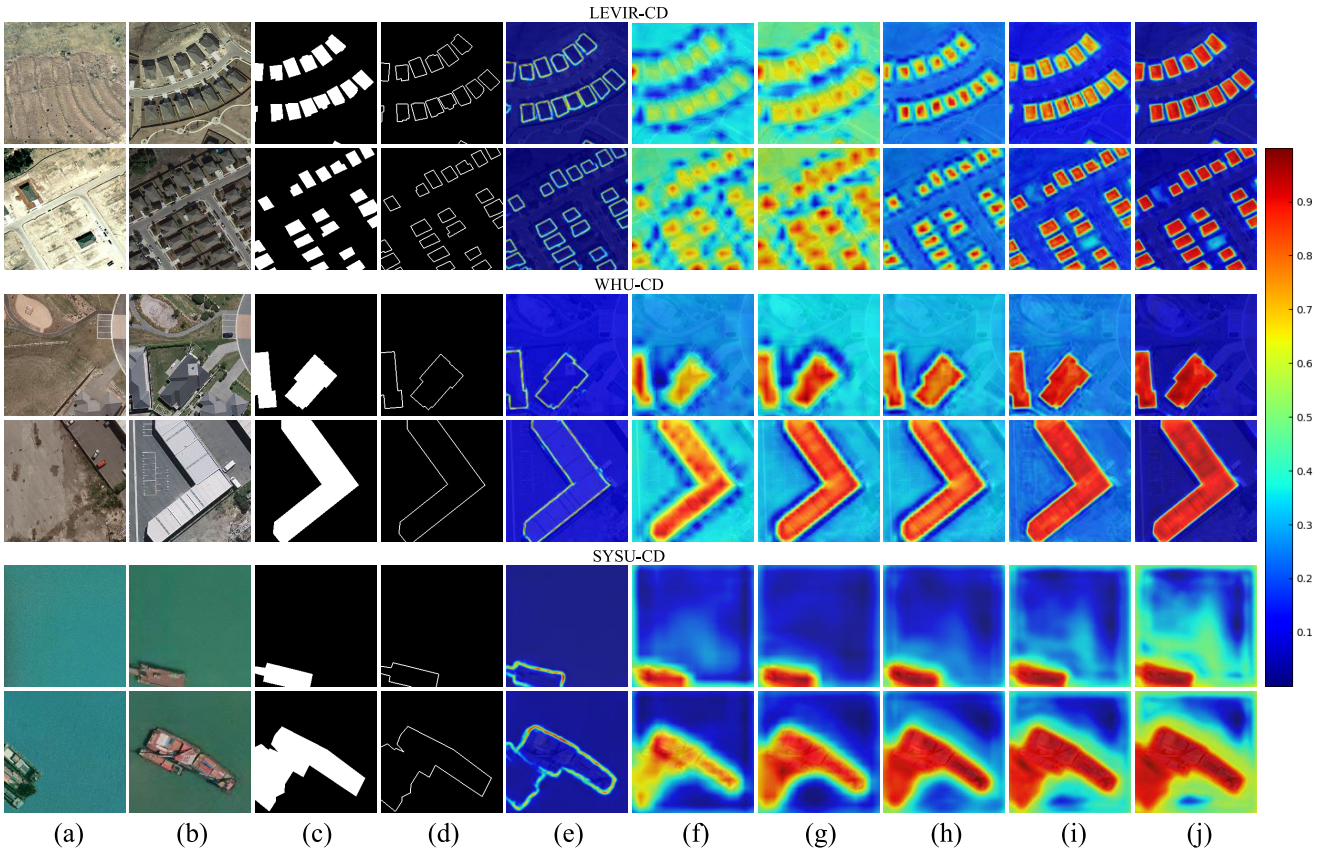


Fig. 10. Visualization results of learning process on three public datasets. (a) $\boldsymbol{T}^1$. (b) $\boldsymbol{T}^2$. (c) $\boldsymbol{GT}$. (d) $\boldsymbol{GT}_e$. (e) $\boldsymbol{EM}$ generated by EEM. (f) $\boldsymbol{CM}_5$ generated by ORM. (g) $\boldsymbol{CM}_4$ generated by EOL at level four. (h) $\boldsymbol{CM}_3$ generated by EOL at level three. (i) $\boldsymbol{CM}_2$ generated by EOL at level two. (j) $\boldsymbol{CM}_1$ generated by EOL at level one.

As shown in #03 of Table II, w/o EEM achieves Kappa, $F1$, and IoU scores of 91.91%, 92.32%, and 85.73% on the LEVIR-CD dataset. These results are lower than those achieved by EOCLNet. Additionally, we further validate the effectiveness of incorporating the highest-level feature $\boldsymbol{F}_4$ in EEM. #04 of Table II provides the results without incorporating $\boldsymbol{F}_4$ in EEM. The results show that not incorporating $\boldsymbol{F}_4$ led to decreases of 0.79%, 0.76%, and 1.4% in Kappa, $F1$, and IoU scores, respectively, on the WHU-CD dataset. #05 of Table II provides the results of using the feature $\boldsymbol{F}_3$ to filter noise interference in EEM. The results show that replacing $\boldsymbol{F}_4$ with $\boldsymbol{F}_3$ in EEM led to decreases of 0.16%, 0.15%, and 0.28% in Kappa, $F1$, and IoU scores, respectively, on the WHU-CD dataset. This confirms that utilizing the highest-level feature $\boldsymbol{F}_4$ in EEM is more effective in enhancing noise robustness.

3) Effectiveness of ORM: The proposed ORM reveals changed objects by aggregating the highest two levels of semantic features. A 1×1 convolution layer is used to replace the ORM (referred to as w/o ORM) to validate its effectiveness. The results of replacing the ORM are reported in #06 of Table II. On the WHU-CD dataset, the Kappa, $F1$, and IoU scores for w/o ORM are 1.40%, 1.35%, and 2.46% lower than those of EOCLNet, respectively. On the SYSU-CD dataset, they are 1.01%, 1.03%, and 1.53% lower, respectively. Table II #07 and #08 also provide the results of using only channel alignment (referred to as w/o spatial alignment) or spatial alignment (referred to as w/o channel alignment) in ORM. The results show that w/o spatial alignment on the LEVIR-CD dataset has Kappa, $F1$, and IoU scores of 92.06%, 92.46%, and 85.98%, respectively, and w/o channel alignment has scores of 92.15%, 92.55%, and 86.13%, respectively, all lower than those of EOCLNet. This validates the effectiveness of combining channel and spatial alignment in ORM. Additionally, we also attempt to aggregate $\boldsymbol{F}_2$ and $\boldsymbol{F}_4$ (referred to as $\{\boldsymbol{F}_2, \boldsymbol{F}_4\}$) in ORM to reveal changed objects, with the results shown in #09 of Table II. It can be observed that $\{\boldsymbol{F}_2, \boldsymbol{F}_4\}$ performs lower than EOCLNet on all three datasets, indicating that aggregating the highest two levels of semantic features, namely $\boldsymbol{F}_3$ and $\boldsymbol{F}_4$, to reveal initial changes is more effective.

4) Effectiveness of EOL: By establishing relationships between edges and changed objects across multiple levels, EOL is used to implicitly mine edge clues. To validate its effectiveness, EOL is replaced with a 1×1 convolutional layer (referred to as w/o EOL) to achieve feature aggregation from high to low levels and output change maps for each level. The results without EOL in the proposed EOCLNet are shown in #10 of Table II. Comparing #01 and #10 from Table II, it can be observed that without EOL, the Kappa, $F1$, and IoU scores decreased by 0.63%, 0.61%, and 1.04% on the LEVIR-CD dataset, 0.91%, 0.87%, and 1.6% on the WHU-CD dataset, and 1.43%, 1.46%, and 2.17% on the SYSU-CD dataset, respectively. Additionally, #11 and #12 of Table II

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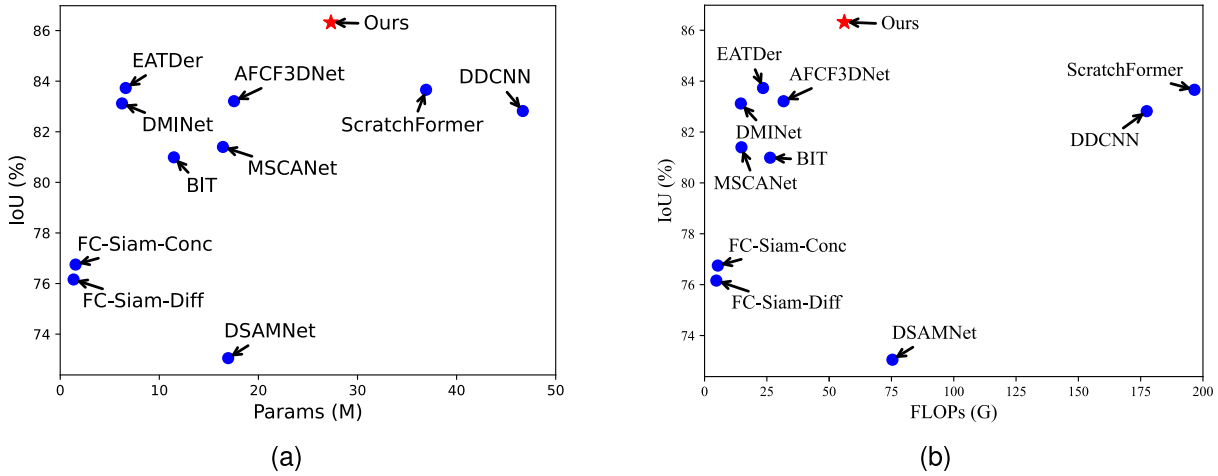


Fig. 11. Model efficiency of different methods. (a) Params versus IoU. (b) FLOPs versus IoU.

provide the results where only the background (referred to as w/o foreground branch) or foreground (referred to as w/o background branch) branch is used in EOL. On the WHU-CD dataset, the Kappa, $F1$, and IoU scores for w/o foreground branch are 0.24%, 0.23%, and 0.43% lower, respectively, than those of EOCLNet, and for w/o background branch, they are lower by 0.45%, 0.44%, and 0.81%. Similar results are observed on the other two datasets, indicating that combining both branches in EOL is necessary.

5) Effectiveness of Ω_m: The uncertainty in pixel attribute determination leads to ambiguity in the CD results. The weight map $\mathbf{\Omega}_m$ is constructed for the BCE loss to impose greater penalties on areas of uncertainty during training, facilitating the transition from uncertainty to certainty. The results without $\mathbf{\Omega}_m$ (referred to as w/o $\mathbf{\Omega}_m$) are shown in #13 of Table II. It can be observed that the performance of w/o $\mathbf{\Omega}_m$ on three public datasets is significantly lower than that of EOCLNet. For example, on the SYSU-CD dataset, it achieves Kappa, $F1$, and IoU scores of 78.95%, 83.81%, and 72.13%, which are 0.9%, 0.84%, and 1.26% lower than those of EOCLNet, respectively. This indicates that increasing attention to areas of uncertainty during training helps improve the model's performance.

G. Visualization of Learning Process

To more intuitively demonstrate the role of edge-object co-driven learning, we visualized the learning processes of EEM, ORM, and EOL on three public datasets, with the results shown in Fig. 10. From Fig. 10(e), it can be observed that the edge regions exhibit higher values, which matches the function of EEM in extracting edge information. ORM is designed to reveal changed objects, as demonstrated in Fig. 10(f), where they are successfully captured. For EOL, it is employed to establish relationships between edges and changed objects across multiple levels to further mine edge clues. This process is validated in Fig. 10(g)–(j), where it can be observed that edge details are progressively enriched.

V. Discussion

This section discusses five aspects: model efficiency, learning curves, training data proportion, parameter settings, and limitations and future work.

A. Model Efficiency

We calculate the Params and FLOPs of different methods using images of size $3 \times 256 \times 256$, with the batch size set to 1 to ensure fairness and accuracy in the comparison. The results are presented in Fig. 11. It can be observed that FC-Siam-Diff and FC-Siam-Conc exhibit lower Params and FLOPs due to their relatively simple structures. However, this simplicity also results in unsatisfactory performance. DDCNN has the highest Params but does not achieve outstanding performance. ScratchFormer has the highest FLOPs, yet its performance is comparable to EATDer, which has significantly lower FLOPs. Moreover, EATDer achieves a good balance among Params, FLOPs, and performance, demonstrating high model efficiency. Our method exhibits superior performance compared to others, while avoiding the high Params and FLOPs of DDCNN and ScratchFormer.

B. Learning Curve Comparison

Fig. 12 displays the learning curves of our method compared to four other methods: ScratchFormer, DDCNN, BIT, and MSCANet. This includes the IoU on the LEVIR-CD validation set [corresponding to Fig. 12(a)] and the CD loss [corresponding to Fig. 12(b)]. It can be observed that our method achieves optimal performance while maintaining good convergence speed and model stability. In contrast, ScratchFormer exhibits weaker model stability. On the other hand, it can also be seen that the CD loss of our method is significantly higher than that of BIT and MSCANet, primarily because we employed multilevel supervised learning to alleviate the learning difficulties of intermediate layers.

C. Training Data Proportion Comparison

To test the performance of the proposed EOCLNet with a small amount of training samples, we set up multiple groups of training data with different proportions. Specifically, in the LEVIR-CD dataset, the training set consists of 7120 image pairs, which account for 70% of the entire dataset. We train the model using 5% (356), 10% (712), 20% (1424), 40% (2848), 60% (4272), 80% (5696), and 100% (7120) of the

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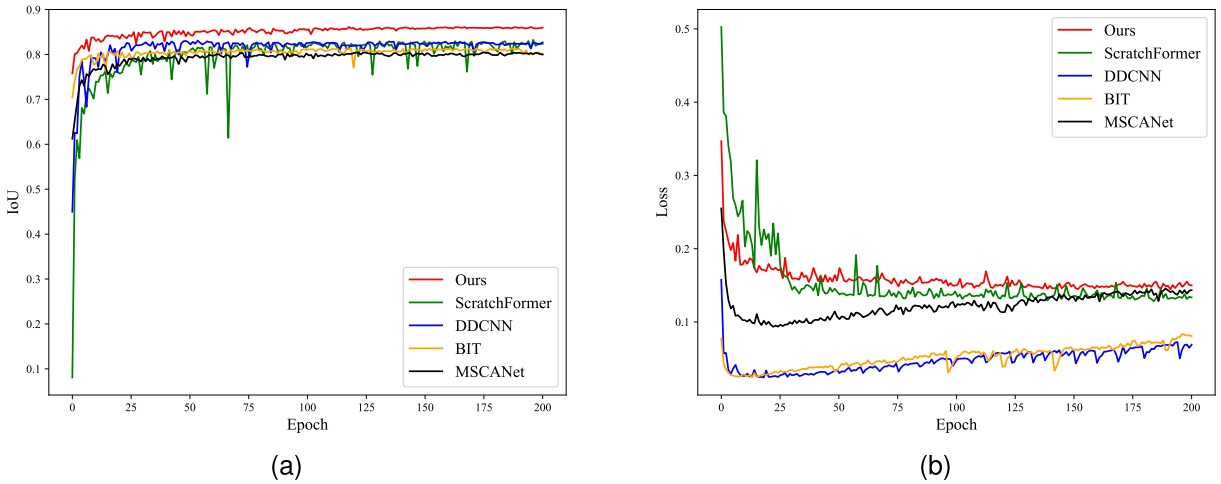
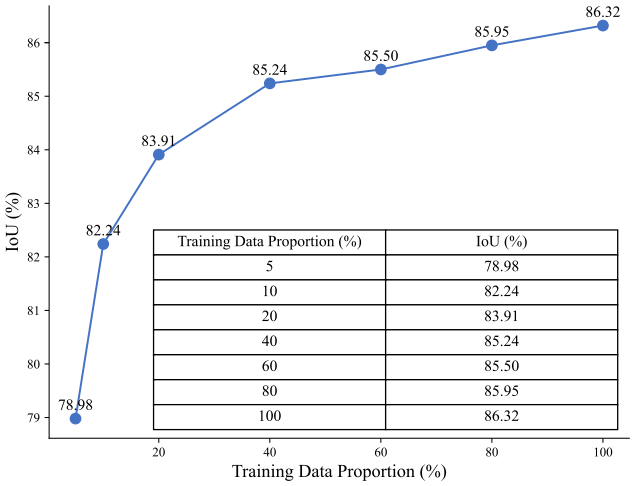


Fig. 12. Learning curves of different methods. (a) IoU on the LEVIR-CD validation set. (b) CD loss on the LEVIR-CD validation set.



Training Data Proportion (%)	IoU (%)
5	78.98
10	82.24
20	83.91
40	85.24
60	85.50
80	85.95
100	86.32

Fig. 13. Results of different training data proportions.

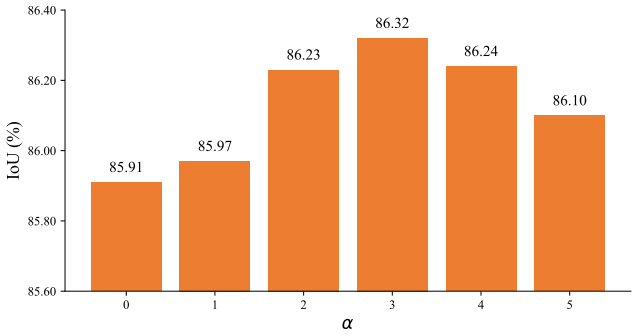


Fig. 14. Results for different α settings.

original training set, while keeping the number of validation and test samples unchanged. The results are shown in Fig. 13. It can be observed that the proposed EOCLNet achieves good performance with relatively few training samples. Specifically, when only 20% of the training data is used, the proposed EOCLNet achieves an IoU of 83.91%, surpassing the 83.73% achieved by EATDer with 100% of the training data. This is attributed to the constructed DPFE, which reduces the reliance on a large number of training samples.

D. Parametric Analysis

Fig. 14 displays the results for different settings of α on the LEVIR-CD dataset, where 0 indicates no added focus on areas of uncertainty, and 1–5 represent progressively increasing penalties on areas of uncertainty during training. It can be observed that the best results come from $\alpha = 3$. When $\alpha > 3$, the model performance begins to decline, though still better than $\alpha = 0$, indicating that excessive penalties are also detrimental. Additionally, a dynamic adjustment strategy for α could be considered. For instance, higher-level change maps with greater ambiguity may benefit from a larger α setting. Conversely, for lower-level change maps, a smaller α setting might be more appropriate. However, for simplicity, this article employs a fixed strategy for α, uniformly setting it to 3.

E. Limitations and Future Work

Although the effectiveness of the proposed method has been validated on three public datasets, it still has some limitations. First, EOCLNet adopts a DPFE, which results in higher Params and FLOPs, thereby affecting the inference speed to some extent. We discuss the model efficiency of different methods in Section V-A. In the future, we could further optimize the feature extraction process. On the other hand, EOCLNet only outputs the changed regions and lacks in-depth semantic understanding of the changes. To provide more comprehensive change descriptions, future work could explore integrating techniques like natural language generation to combine semantic understanding with CD.

VI. Conclusion

This article proposes a CD network called EOCLNet, aimed at addressing insufficient edge detail recognition and limited feature extraction. It utilizes FastSAM to obtain a broad

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range of general features to compensate for the inadequacies of features extracted from limited data, which is achieved through the constructed DPFE. For edge detail recognition, it includes the explicit learning of EEM and the implicit mining process of EOL. ORM provides initial change guidance by aggregating the highest two levels of semantic features. Additionally, during the training process, the uncertainty in the previous level's change map is used to guide the learning at the next level, achieving a transition from uncertainty to certainty. A series of experiments conducted on three public datasets validate the effectiveness of the proposed EOCLNet, which outperforms ten comparison methods, effectively preserving edge details.

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